

6.1 Sourcing Open Data

Data Source:

The dataset is sourced from Kaggle, but comes from the book *Machine Learning with R* by Brett Lantz, which provides an introduction to machine learning using R. Although the dataset is fictional, it will be used to for this project explore factors that contribute to higher medical costs. The dataset provides 1,338 records of medical charges for adults 18-64. Each Entry contains information about each insurance contractor including factors that could lead to higher medical charges.

The data can be accessed here:

<https://www.kaggle.com/datasets/mirichoi0218/insurance>

Data Collection:

The dataset being used was pulled from the book *Machine Learning with R* by Brett Lantz. It is a synthetic dataset which is a collection of artificial data that is generated to mimic the characteristics and patterns of real-world data, essentially creating a "fake" dataset that shares similar statistical properties with the original data, but without containing any actual sensitive information from the real data source.

The dataset aggregates information about each insurance contractor including: age, gender, bmi (body mass index), number of dependent children, smoking status, region, and medical charges.

Data Limitations:

1. Limited Variables – The data shows factors that could contribute to higher medical charges for insurance contractors. However, many more variables could contribute to medical charges that are not part of the dataset.
2. Lack of Context – The data does not give specific years when the data was collected and therefore might lack relevance for the present.
3. Synthetic Dataset – although created to mimic the characteristics and patterns of real-world data, the data does not represent actual insurance contractors.

Why This Data?

I choose this data because I liked the idea of exploring the contributing factors of each insurance contractor to their corresponding medical charges. Although a synthetic dataset, I liked the idea of analysis relationships between the variables outside of the medical charges. Such as in what region contain the most smokers or highest number of individuals with a high BMI.

I felt this would dataset would allow me to utilize some of the skills I used in this program not to mention explore information within healthcare, an industry I have worked in.

Ethical Considerations:

Due to the dataset being synthetic, any insights derived should not be used to make real-life conclusions. All insights from this data, will be used solely for this project to show case data analytic skills learned.

Questions to Explore:

What regions contain the highest number of smokers?

What regions contain the highest number of those with an above average BMI?

How many insurance contractors who smoke have child dependents?

What factors contribute the most to high medical costs?

Data Cleaning and Consistency Checks

Checked for mixed data types – non found

Checked for missing values – non found

Checked for duplicates – one duplicate found and deleted

Data Profile

Column Name	Description	Variable Type
age	Age of insurance contractor	int64
sex	Gender of insurance contractor	object
bmi	Body Mass Index (BMI) is a calculation used to estimate body fat percentage based on a person's height and weight.	float64
children	Number of children dependents on insurance plan	int64
smoker	Smoking status of insurance contractor	object
region	US region where insurance contractor resides	object
charges	Medical charges associated with insurance contractor	float64

Statistical Analysis

	age	bmi	children	charges
count	1337.000000	1337.000000	1337.000000	1337.000000
mean	39.222139	30.663452	1.095737	13279.121487
std	14.044333	6.100468	1.205571	12110.359656
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.290000	0.000000	4746.344000
50%	39.000000	30.400000	1.000000	9386.161300
75%	51.000000	34.700000	2.000000	16657.717450
max	64.000000	53.130000	5.000000	63770.428010